Online Banking Enterprise™

Payments – Bankserv (ACB) File Format »

If you are using the standard Bankserv (ACB) file layout you will be able to import a file containing multiple execution dates as well as multiple From Account numbers i.e. you can pay recipients on different days and use different accounts to pay recipients.

Online Banking Enterprise ™ will create a batch for every contra record in the imported file. For example: you can create a batch called Creditors and import a file with three contra records, three batches will be created on Online Banking Enterprise ™ with the following names; Creditors00001, Creditors00002, Creditors00003. This is only available for the Bankserv (ACB) file format.

If you do not have an action date in the contra record field in their file, the system will automatically default the date to the date of the import. To future date the file, the action date MUST be reflected in the Contra record of the file.

- The maximum number of payments which can be contained in the import file is 100,000.
- The maximum number of contras which can be contained within a single file is 200.

The import file used by Online Banking is the standard Bankserv (ACB) file format. The record length is 181 characters, including the Carriage Return (ASCII 13) and Line Feed (ASCII 10) control characters.

INSTALLATION HEADER RECORD

02000000000

STANDARD TRANSACTION RECORD

1025065562010091234000100000225065562011234567100000000001001109070000USER REFERENCE HOMING ACCOUNT 00000000000000000000

Field	Field Name	Length	Character Position	Description	Alpha or Numeric	Example of Record Content
1	Record Identifier	2	1-2	• Must contain the two- digit code identifying the type of transaction: • Use 10 or 20 for Payments	N	10
2	User Branch	6	3-8	Must contain the valid Branch code identifying the branch of the Bank of your Nominated Account.	N	250645
3	User Nominated Account Number	11	9-19	Must contain your valid Nominated Account number.	N	0200076260
4	Filler	4	20-23	Must contain blank spaces or zeros.	AN	0000
5	User Sequence Number	6	24-29	Must be in ascending sequential order.	N	000001

Field	Field Name	Length	Character Position	Description	Alpha or Numeric	Example of Record Content
6	Homing (Recipient) Branch	6	30-35	Must contain the valid code identifying the branch of the Bank of your Recipient's Homing Account. The universal branch codes for the applicable bank will be accepted. **Notes:** • When importing a Public Recipient, enter six zeros; i.e. 000000. • When making payments to eWallet recipients, please use the FNB Universal Branch Code: 250655.	N	632005
7	Homing (Recipient) Account Number	11	36-46	Must contain a valid Homing Account Number for your Recipient. **Notes:** Zero fil this field and use the non-standard account number (field 18) when making payments to Public Recipients and eWallet.	N	00710423598

Field	Field Name	Length	Character Position	Description	Alpha or Numeric	Example of Record Content
8	Type of Account	1	47-47	Must contain a valid Account type for the Recipient's Homing Account. The following type of account codes apply: **Account Type** / **Value** Public Recipient: 0 Current (cheque/bond) Account: 1 Savings Account: 2 Transmission Account: 3 Bond Account: 4 Subscription Share Account: 6 eWallet Account (eWallet Pro): D eWallet Account (Send Money): S	N	1

Field	Field Name	Length	Character Position	Description	Alpha or Numeric	Example of Record Content
9	Amount	11	48-58	• Must contain the value of the transaction which must be greater than zero. • The value must not exceed the item limit, either apart from, or when combined with other transactions for the same account for the same Action Date, for the same User Data Set. • Amount must be expressed in cents without using any currency indicators, comma and/or decimal point separators. For example, R3,728.68 must be captured as 00000372868 in the amount field.	N	00000372868
10	Filler	8	59-66	Must contain zeros.	N	00000000
11	Tax Code	1	67-67	Should always be zero.	N	0
12	Filler	2	68-69	Must contain zeros.	N	00
13	Filler	1	70-70	Must contain zero.	N	0

Field	Field Name	Length	Character Position	Description	Alpha or Numeric	Example of Record Content
14	To Account Reference	20	71-90	Must contain a suitable explanation of the transaction. This field is used as part of the Recipient reference and is printed on the recipient's bank statement. • **Payments**: 20 characters used for the To Account Reference.	AN	USER REFERENCE
15	Filler	10	91-100	Must contain blank spaces or zeros.	AN	0000000000
16	Homing (Recipient) Account Name	15	101-115	Must contain the name of the Homing (Recipient) Account holder. This field is used as part of the Recipient Name and Own reference on the system. **Note:** These names serve as reference only and are not validated against the account number when the payment takes place.	AN	HOMING ACCOUNT
17	Filler	15	116-130	Must contain blank spaces or zeros.	AN	000000000000000

Field	Field Name	Length	Character Position	Description	Alpha or Numeric	Example of Record Content
18	Non-Standard Account Number	20	131-150	This field serves to accommodate account numbers, which cannot be held in field 7 above, i.e. account numbers longer than 11 digits. **This may be due to the following:** • The Homing (Recipient's) Account number is longer than 11 digits • Public Recipients – enter the reference provided for the recipient on the Public Recipient List. • Send money to eWallet – enter the recipient's 10-digit South African cellphone number. • eWallet Pro – enter the eWallet account number for the recipient. **Note:** • If this field is used, the homing account field 7 must be zero filled. • If this field is not used it must be zero filled.	N	00000000000000000000
19	Filler	30	151-180	Must contain blank spaces or zeros.	AN	000000000000000000000000000000

CONTRA RECORD

A contra record is required for each action date and/or From (nominated) Account in the file, a separate contra record (Record 12) must be created for each dataset.

A maximum of 200 contras can be entered in a single file.

1225065562011234567000000000000000000000000000000000000000000YYMMDD000000000000000000000000000000000000000000000000000000000000000000000000
0

Field	Field Name	Length	Character Position	Description	Alpha or Numeric	Example of Record Content
1	Record Identifier	2	1-2	Must contain the code which accurately identifies the contra type: • If you use Record Id 10 in your Standard transaction record Identifier field, then you must use 12. • If you use Record Id 20 in your Standard transaction record Identifier field, then you must use 22.	N	12 or 22 (depending on Record Identifier used in the Standard Transaction Record)
2	User Branch code	6	3-8	This is an Optional field - Required when using Multiple From (nominated) account numbers in the file. • If used - Must contain the valid Branch code identifying the branch of the Bank of your Nominated Account. **Note:** If no from Account numbers are given, the system will use the details captured at the time of creating the batch header details within Online Banking Enterprise™.	N	250655

Field	Field Name	Length	Character Position	Description	Alpha or Numeric	Example of Record Content
3	User Nominated Account Number	11	9-19	This is an Optional field - Required when using Multiple From (nominated) account numbers in the file. • If used - Must contain your valid Nominated Account number. (can differ according to transaction set) **Note:** If no From Account numbers are given, the system will use the details captured at the time of creating the batch header details within Online Banking Enterprise™.	N	62011234567
4	Filler	39	20-58	Must contain blank spaces or zeros.	AN	
5	Action date	6	59-64	This is an Optional field – if no Action date is provided, the system will automatically default the date to the same date of the import. **Important**! If you want to make use of the Future date functionality, you need to include the required Action date in this field.	N	Format = YYMMDD
6	Filler	116	65-180	Must contain blank spaces or zeros.	AN	

USER TRAILER RECORD

92000000000

INSTALLATION TRAILER RECORD

9400000000000

Hash Totals and Duplicate Batch Checks on Bankserv (ACB) and CSV import files »

Hash Totals Overview

Your Administrator(s) can enable the Hash Totals and Batch Duplicate Check functionality on your Online Banking Enterprise™ profile via Site Settings.

A Hash Total is an optional security layer on Online Banking Enterprise™ which is used to verify the accuracy of data in a file by using an algorithm. In addition to the Hash Total validation on File Import, you may also select to perform a Batch Duplication check.

How does Hash Totals work?

Your line of business application calculates the Hash Total value for the Payment or Collection file and populates it in the respective field as per the ACB or CSV file specification. When you import your file into Online Banking Enterprise™, our system will calculate the Hash Total value and compare it against the Hash Total value in your file.

If these values do not match, the file will fail and will not be imported into Online Banking Enterprise™. Enabling this functionality helps prevent unauthorised editing of Payments and Collection import files.

Important Notes:

- The Administrator(s) must enable the Hash Total functionality via the Profile Site Settings.
- The Administrator(s) must enable these settings for each entity if they have a hierarchy structure. The Settings are only related to the Entity that was selected and not to other Entities in the Hierarchy. This means that the Hash Total Validation is only done on Entities for which it is enabled.
- Hash Total Validation is done on both Bankserv (ACB) and CSV file imports.
- **All Payments and Collections files must be single contra batch files when using Bankserv (ACB) file imports.**
- The Hash Total Information of a batch is used for comparison if the batch status is not Work In Progress.

- If the Entity is enabled for Hash Total use and the Hash Total validation on Online Banking Enterprise™ fails, the Hash Total Error Message will be displayed on the Print Preview page.

How is a Hash Total calculated on a Bankserv (ACB) File?

- Add the 11 digits of the **To Account Numbers** (recipient) and the **Non-Standard Account Number** (First 11 digits only from the right) of each standard transaction record.
- Once the total sum of the accounts in point 1 has been determined, add the 11 digits **From Account Number** only **once** to the total of point 1.
- Use the **last 12 digits** (Right justified) of the Total of point 1 and 2. E.g. 12345**678987654321.** These last 12 digits from the right will be used to populate the User Trailer Record. **Place the digits in record 92 from position 73 to 84**.

Batch Duplication Checks

The **Perform Duplicate Batch Check** function can be selected with or without selecting the Use Hash Total Validation on File Import (data integrity checks on imported batches) option. The Batch Duplication function reduces the risk of duplicating imported or captured batches on Online Banking Enterprise™ by warning you if the file being imported is a duplicate of a file already loaded into Online Banking Enterprise™ or of batch transactions that were captured on Online Banking Enterprise™.

This function will check and confirm if the same **Batch Name and/or Batch Content** (Homing Account Numbers, References and Amounts) is detected for a Recipient on a captured or imported batch that already exists within another batch on Online Banking Enterprise™ and will action the request based on the option selected for **Action on Duplicate**.

The following options can be selected from the **Action on Duplicate** drop-down list:

- **Continue with Batch**
  - The system will NOT warn you if it detects any duplicated details and will allow you to submit batches as normal.

- **Warn User**
  - The system will detect that you are submitting the same Batch Name and/or Batch Content (Homing Account Numbers, References and Amounts) for a Recipient which already exists on another captured or imported batch on Online Banking Enterprise™ and will allow you to Cancel or Submit the batch.

- **Prevent Submission of the Batch**
  - The system will detect that you are submitting the same Batch Name and/or Batch Content (Homing Account Numbers, References and Amounts) for a Recipient which already exists on another captured or imported batch on Online Banking Enterprise™ and will not allow you to proceed with the submission of the batch.

Hash Total Duplication Checks

This functionality verifies that the **Hash Total for the batch** being processed **is unique** and does not resemble the Hash Total Information of another batch that was successfully captured previously.

- Hash Total Duplication is only done on Entities for which the Perform Batch Duplicate Check option is enabled and the action option on a duplicate response is set to **Warn User** or **Prevent Submission of the Batch**.
- The Hash Total Duplication check is done on both Payments and Collections batches.
- If a batch is rejected, its Hash Total Information is removed from the Hash Total List to prevent it being used to evaluate other batches against.
- Batches will only have their Hash Total Information logged and the duplicate check will be performed against historical previous batches that were captured or imported when the Perform Batch Duplicate Check setting was enabled for the entity. Batches captured or imported when the functionality was switched off will not have Hash Total information stored and no duplicate checking will be done.
- The Hash Total used for Batch Duplication checking is stored together with the Batch Name and Date of Submission and is kept in accordance with the entity's Payments and Collections Retention Days settings even if the user deletes the batch. It is not removed if the Entity is deactivated for Hash Total Use.

How to enable the Hash Totals and Duplicate Batch checks on your Online Banking Enterprise™ profile.

The Administrator(s) must **enable the Hash Total functionality** and select if they want to perform **Duplicate Batch checks** via Profiles > Site Settings > Hash Totals **>** Perform Batch Duplicate Check.

MD5 and SHA1 Hash Strings validation on Bankserv (ACB), CSV and ISO import files »

MD5 and SHA1 Hash Algorithms Overview

These Hash Algorithms are used to facilitate secure digital transactions in your Line of Business Systems

The MD5 and SHA1 secure Hash Algorithms were developed by the National Institutes of Standards and Technology (NIST) and other government and private parties.

Important Notes:

Please note that the MD5 and SHA1 Hash Algorithms checks work independently of the Hash Totals and the Duplicate Batch Check functionality enabled via the Site Settings on your Online Banking Enterprise™ profile.

How do the MD5 and SHA1 algorithms work?

If you are using hash string generating software to calculate the hash values for the Payment and/or Collections files that you generate via your Line of Business systems, you can compare these hash strings with the MD5 and SHA1 hash strings that we display on Online Banking Enterprise™ at the time of your file import. If the hash strings match, this confirms that the data in the respective file has not been tampered with.

MD5 and SHA1 hash strings are **automatically displayed on all Payment and/or Collections files imported on Online Banking Enterprise™;** therefore, these Hash Algorithms essentially act as a "file check" function to help facilitate secure digital transactions on your private network.